

Chapter 02: Software Requirement – Testing Requirement

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- Software requirement
- Requirement classification
- Requirement clarifying
- Requirement modeling
- Common issues and solutions
- Test Requirement

Software requirement

Software requirement definition #1

- Software development requirements refers to the set of **specific**, **measurable** and **testable** characteristics that a software system must have to meet the needs of its users.
- A statement of a **service** the system must do **OR**
 - Example: the user shall be able to add a new contact to the address book
- A statement of a **constraint** the system must satisfy
 - Example: a search on contacts shall return a result in no more than 2 seconds

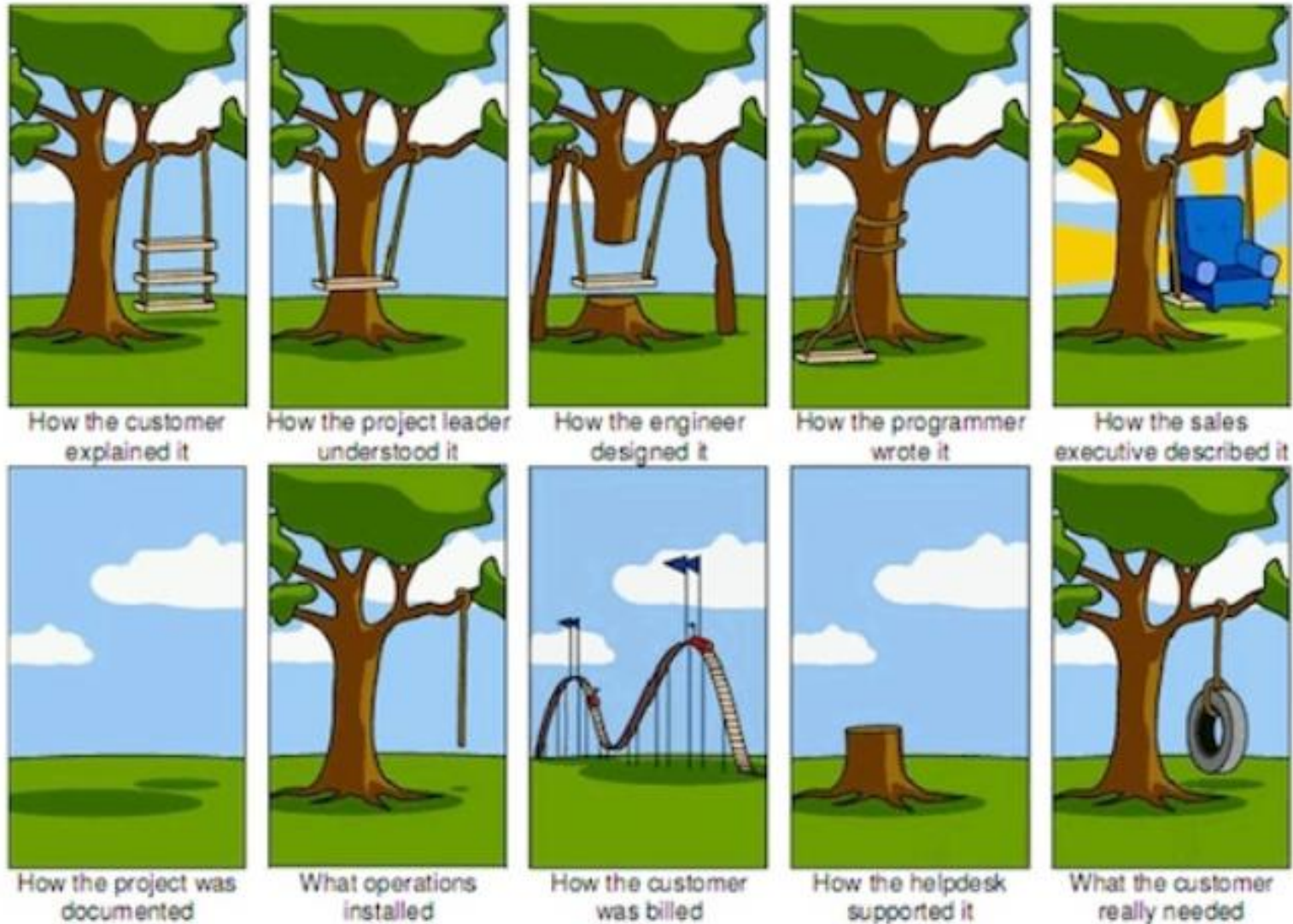
Software requirement definition #2

- As defined by the IEEE Standard Glossary of Software Engineering Terminology [1] a requirement is:
 1. A condition or capability needed by a user to solve a problem or achieve an objective.
 2. A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document.
 3. A documented representation of a condition or capability as in 1 or 2.

What are not Software Requirements?

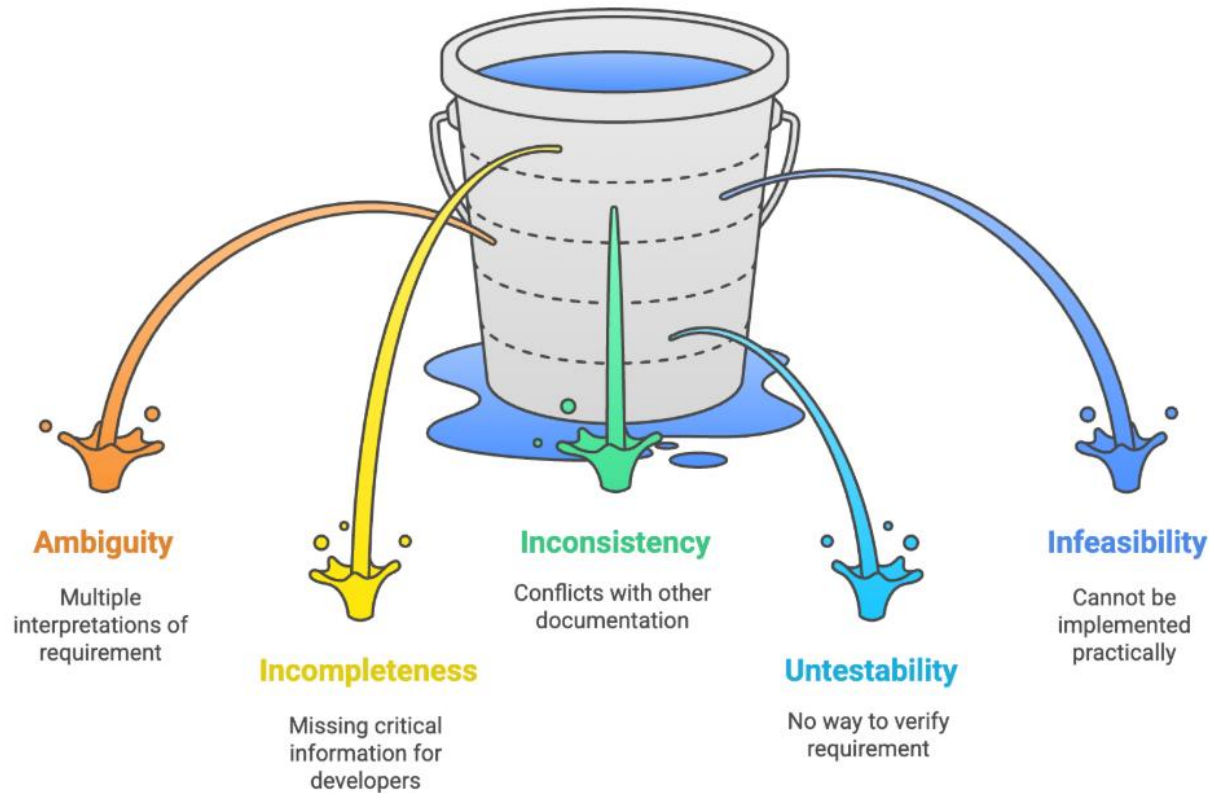
- Design or implementation details (other than known constraints)
- Project planning information
- Testing information

Why do we need requirements? #1



Why do we need requirements? #2

- Always spend time gathering, analyzing and documenting software requirements.

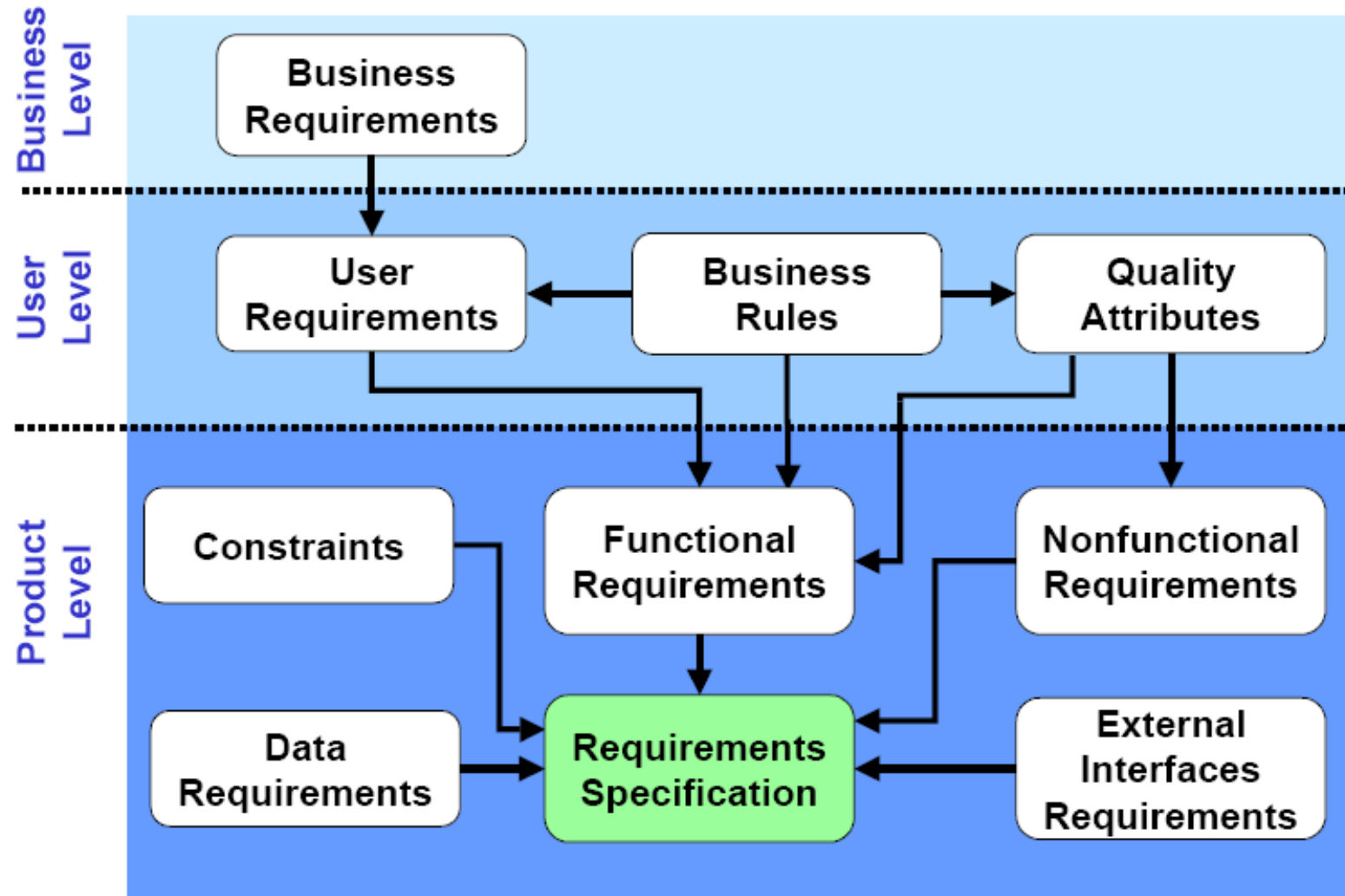


Purpose of requirement

- Requirements often serve as:
 - The basis for a bid for a contract - therefore must be high-level to open for interpretation
 - The basis for the contract itself - therefore must be detailed
- Requirements can be high-level or detailed.

Requirement classification

Requirements Classification



Requirements Classification #2

- Example:
- Business level: "Increase market share by 20% in 6 months"
- User level: "As a salesperson, I need to be able to see inventory in real time."
- Product level: "The system must support login via Google and Facebook."

Requirements Classification #3

- Example:
- Business Rules (business rules related to roles): "Loyal customers receive a 10% discount on orders over 1 million VND."
- Quality Attributes:
 - "Fast lookup" → specify "search under 2 seconds"
 - "The system must handle 1000 simultaneous users"

Requirements Classification #4

- Example:
- Constraints:
 - "The system must use a legacy database specified by the customer"
- Data Requirements
 - "Transaction history must be saved for 5 years"
- External Interfaces Requirements
 - "The system must support the Epson TM-T88 printer"
 - "The software must be able to connect to the partner bank's API"

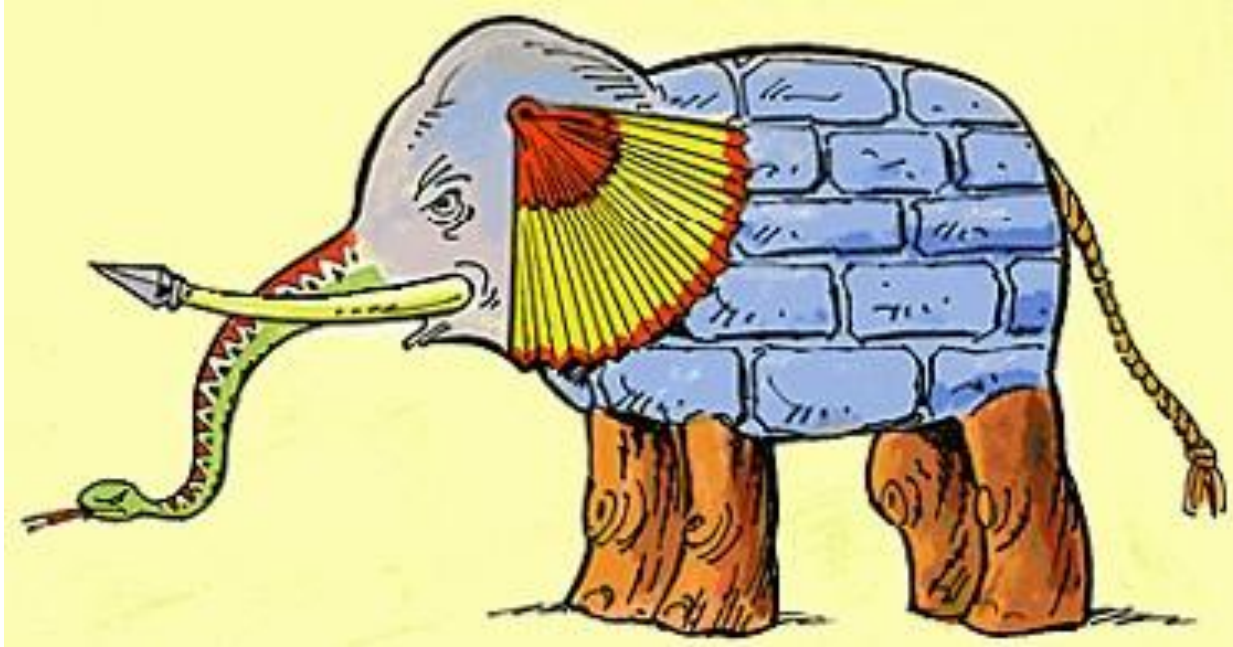
Requirement clarifying

Requirement clarifying

1. PM/TL/BA present the SRS to members in the team
2. Self study related materials: top-down approach
3. Using SRS review checklist
4. Clarify unclear item(s) using Q&A
5. Discuss with other members to clarify or confirm your understanding
6. Media: direct discussion or via team brainstorming
7. Inform the PM/TL/BA about any requirement conflicts changes, comparing to the last version

Clarifying requirements = Reading + Asking + Discussing + Confirming + Notifying

Why do we need Q&A? #1



BA thinks about business processes.
Developers think about technical aspects.
Testers think about test scenarios.
Customers think about business benefits.
Each party only sees a part of the "elephant" (the system).

Why do we need Q&A? #2

Problem	Example
Ambiguous language	"The system must be fast" → How fast are we talking about in seconds?
Unspoken assumption	Everyone thinks, "Of course, there must be a login" but the documentation doesn't mention it.
Differences in knowledge domains	BA writes: "Remove customer from mailing list" In BA terms: stop sending them emails. But DEV understands it as potentially permanently deleting them from the list, losing all their email history.
Missing information	The document says "send email" but doesn't specify when to send it or to whom.
...	...

How to make Q&A effectively?

- Identify the issue: unclear, get for more information, etc.
- Check in all documents that customer supplied to make sure your question has not solved;
- With technical question, check your team/group/company or search Google to solve it before asking out
- Give the cross-reference clearly, completely
- Attach sample screen, demo, give your suggestions if any
- Convert questions to Y/N or multiple-choice types if possible
- In Q&A, give deadline that you want to receive the answer. If there is no answer until the deadline, what is the impact?
- Take the receiver to re-read the question before sending

Requirement modeling

Requirement modeling #1

- Why model requirement?
 - To understand clearly the functionalities of system
 - To present the system from different perspectives
- System modelling helps the analyst to understand the functionality of the system and models are used to communicate with customers.

Requirement modeling #2



Common issues

- When a requirement is unclear, do not rely on assumptions about the customer's intent.
- Requirement analysis is not documented/centralized recorded -> misunderstanding
- How to verify the understanding of team members about requirement?

What should we do

- Maintain SRS
- Communicate with the customer, e.g: online meeting, onsite...
- Don't make assumption. Must confirm with customer about specs more and more about what are still not clear or too general.
- Should involve all members to investigate requirements from the beginning of the project, not only Project Lead do that phase.
- Conduct meeting to verify the requirement understanding.
- One meeting to introduce the requirement/design.
- Another meeting to verified the understanding of team member. Every one have to present their understanding.

Test requirement

Test requirement

- A **test requirement** is a condition or capability needed to verify a software function or quality attribute.
- It defines what to test, how to test and expected results.
- Key distinction
 - Requirement → what the system must do.
 - Test Requirement → how to confirm the system does it correctly.

Requirement	Test requirement
User can log in with email and password	Verify login with valid email/password → success
Search result returns in ≤ 2 seconds	Measure response time for 100 searches → $\max \leq 2$ sec
...	...

Key elements of a test requirement #1

- A complete test requirement must include:

Element	Description
ID	Unique identifier (e.g., TR-LOGIN-01)
Description	What is being tested (brief and clear)
Preconditions	Conditions that must be true before testing
Input Data	Specific values or test data to use
Expected Result	What should happen after the test
Pass/Fail Criteria	How to determine success or failure

Key elements of a test requirement #2

- Example

Element	Description
ID	TR-LOGIN-01
Description	Verify login with correct email/password
Preconditions	User account exists, system is running
Input Data	Email: "test@example.com", Password: "Pass@123"
Expected Result	User is redirected to dashboard
Pass/Fail Criteria	Pass if dashboard appears; Fail if error shown

Examples & Tips

Bad (unclear)	Good (clear)
Check login works	Verify login with valid email (test@mail.com) and password (Abc@123) → dashboard appears in ≤ 3 seconds
Search is fast	Measure search response time for 50 concurrent users → 95% of requests return ≤ 1.5 sec
System handles errors	Verify that entering wrong password 3 times → account is locked for 15 minutes
...	...

Common mistakes to avoid

Mistake	Fix
Vague words: "fast", "works", "properly"	Use specific numbers, actions, outcomes
Missing preconditions	Always state initial state
No pass/fail criteria	Define exactly what Pass means
Mixing multiple tests in one	Split into separate Test Requirements

Quick checklist

- Can a tester execute this without asking for clarification?
- Is the expected result measurable/observable?
- Are input values specific?
- Is there exactly one way to pass?

References

- [1] <https://standards.ieee.org/ieee/729/967/>

Q & A – Thank you